Class 7 : Risk and Risk Aversion

Finance in the morning

- Is Greenland risk?
- Is it risk or uncertainty?

Risk and Uncertainty

- Dollar and US stocks fall as Trump says ‘no going back’ on Greenland bid
  - Stocks off about 2%.
  - Bond prices fell (yields went up)
  - Dollar down from .86 to .85 Euros
  - Bitcoin down 4%.
- Risk as volatility that is quantifiable ex-ante
- Uncertainty is volatility that is not quantifiable ex ante

BEM 103

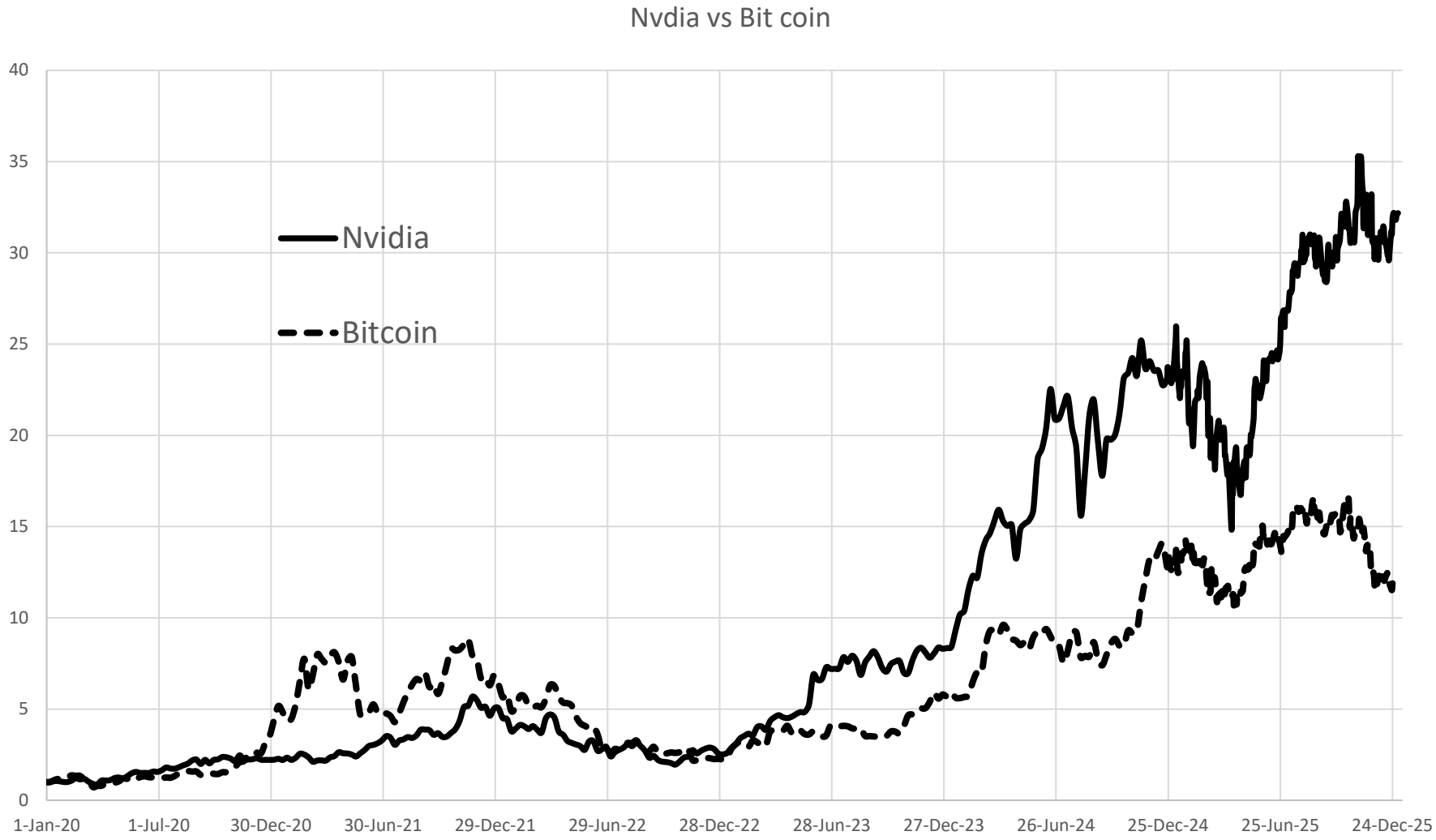
Class 7 : Risk and Risk Aversion

- Risk
- Risk and return
- Risk aversion
- Implications for asset pricing

Key concepts

- States of the world
- Risk aversion
- Lottery
- Expected value
- Expected utility
- Utility of the expected value
- Certainty equivalence
- Insurance

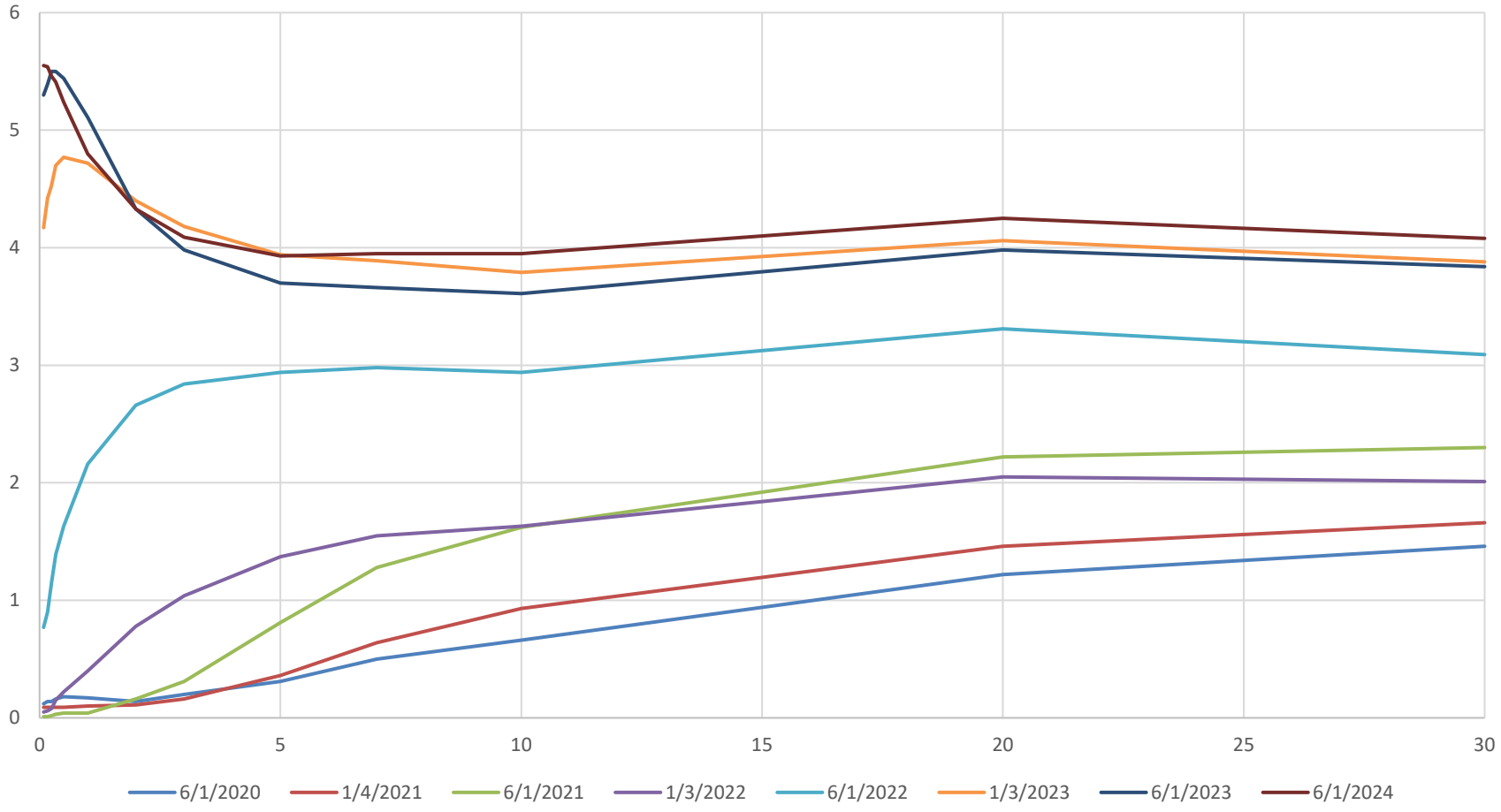
Risk is here (Class 01)



And Here

Index performance 1985-2023
2,000.00
1,800.00
1,600.00
1,400.00
1,200.00
1,000.00
800.00
600.00
400.00
200.00
0.00
S&P500
Nikkei 225
DAX
January-85
January-90
January-95
January-00
January-05
January-10
January-15
January-20
January-25

And here (Class 06)
Yield Curves



First Rep. Bank historical statistics

	return	Variance	Coef of Var
2014	1.24%	4.32	4.8%
2015	24.66%	17.49	8.1%
2016	38.78%	39.78	10.3%
2017	-1.30%	13.40	4.5%
2018	-4.13%	18.55	5.3%
2019	35.21%	36.88	7.1%
2020	25.52%	142.05	12.5%
2021	40.03%	269.94	10.3%
2022	-40.98%	343.81	14.5%
2023	-99.97%	1808.76	186.6%
2024	-97.30%	0.00	92.2%

Both return and variance evolve over time

Some of that is forecastable
Some of that is not

Some of what matters most is not
WHY?

- Assets have variable returns
  - Some of this variation is quantifiable—Risk
    - The return has an expected value and a variance (and may be other moments of interest)
    - Daily variance in the price of
  - Some is not => Risk vs Uncertainty

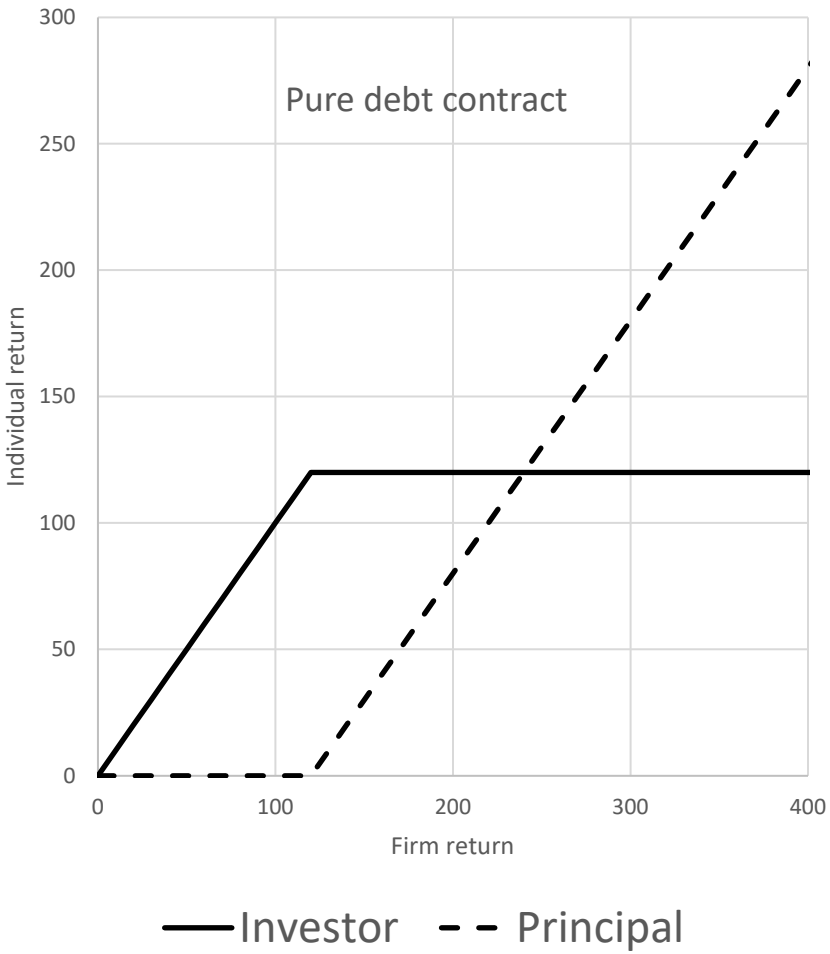
Risk neutrality (Review)

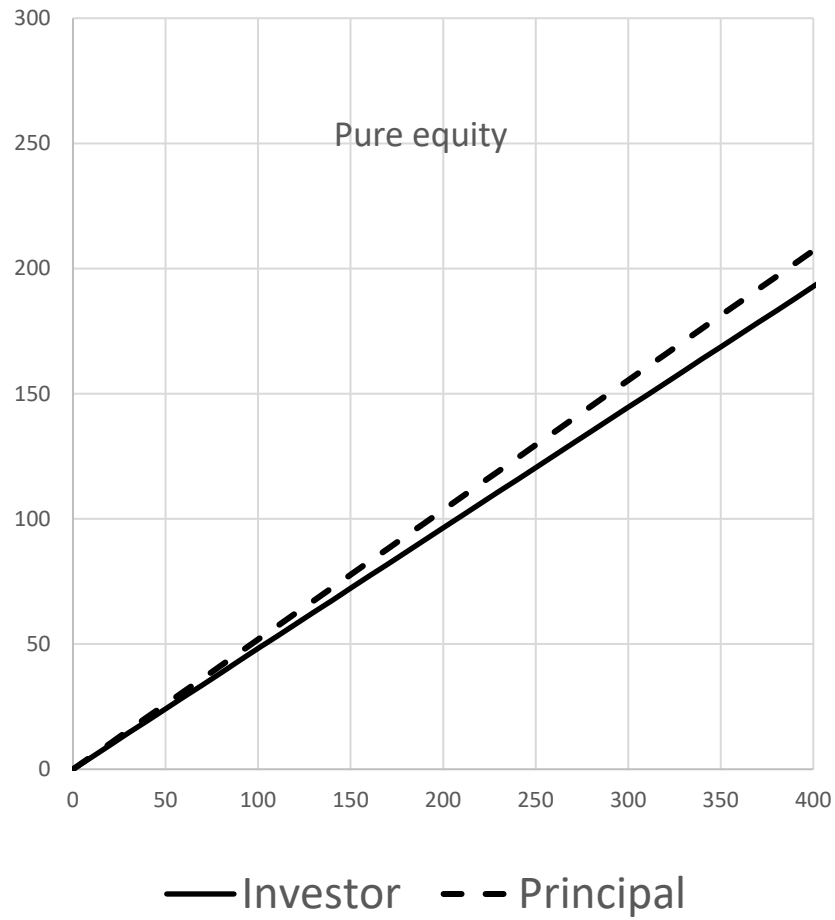
- Risk neutrality
  - Holder of the asset/security only cares about the expected value.
  - So the price of the security today is the discounted expected value (DEV).
  - DEV is something we can compute if most of the future variation in the asset price is risk
    - For long positions, risk will wash out of the long-term trend
    - For leveraged positions, things are different
- Assume all the future variation is risk
  - Bad assumption in the real world
    - Venezuela and Greenland can happen

Implications of Risk

- If the world is certain
  - Entrepreneur needs V to start a project that yields $\prod$ in 1 period
  - Entrepreneur is indifferent between debt and equity.
  - Debt borrow V and repay B=V(1+r) at end
    - Note only lend if $\prod > B$, Entrepreneur return is $\prod - V(1+r)$.
  - Equity raise V for a share α of the return (Investor gets $\alpha\prod$)
    - If market is competitive $\alpha\prod = V(1+r)$.
    - Entrepreneur return $(1-\alpha)\prod = \prod - V(1+r)$.
    - Note if $\prod = B$ then $\alpha=1$
- If the world is uncertain then debt and equity are not the same.

Debt and equity





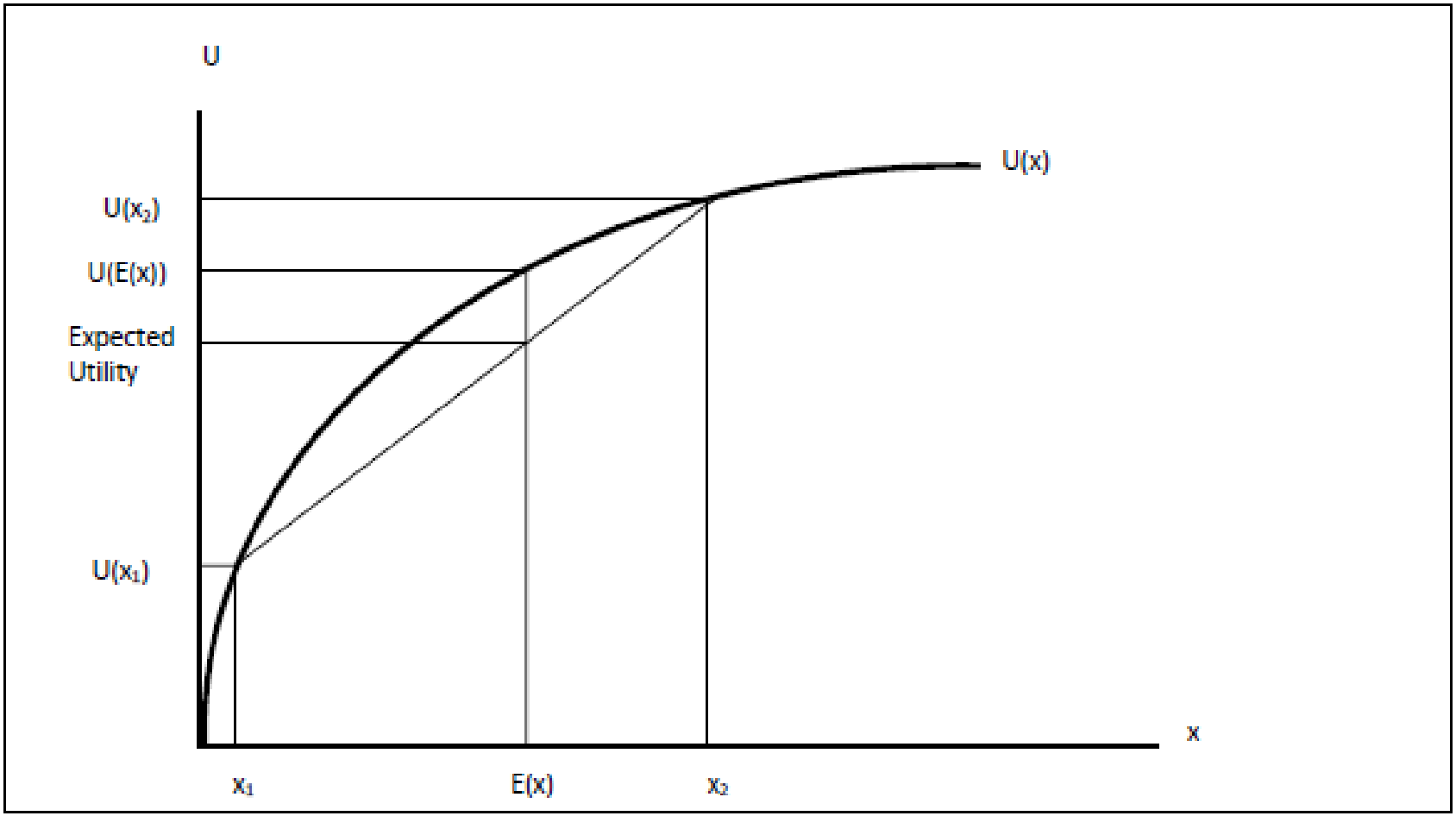
The graphs are after the realization of the uncertainty/risk
At the capital raising stage if both entrepreneur and investor are risk neutral $\alpha EV(\prod) = EV(\min(\prod, V(1+r))$. And the entrepreneur and the investor do not care about what is chosen. Both like more $\prod$ but in different ways.

Risk aversion (back to Ec11)

- About preferences—thus about utility function or the structure of payoffs (not money)
- Ignore the price of time. Suppose there is a lottery that pays
  - H with probability p and L with probability 1-p.
  - Expected value is $pH + (1-p)L$
  - Utility of the expected value $U(pH + (1-p)L)$
  - Expected Utility is $pU(H)+(1-p)U(L)$
- Risk neutral
  - $U(pH + (1-p)L) = pU(H)+(1-p)U(L)$
- Risk averse
  - $U(pH + (1-p)L) > pU(H)+(1-p)U(L)$
- Risk loving
  - $U(pH + (1-p)L) < pU(H)+(1-p)U(L)$

Risk aversion in Graphic terms

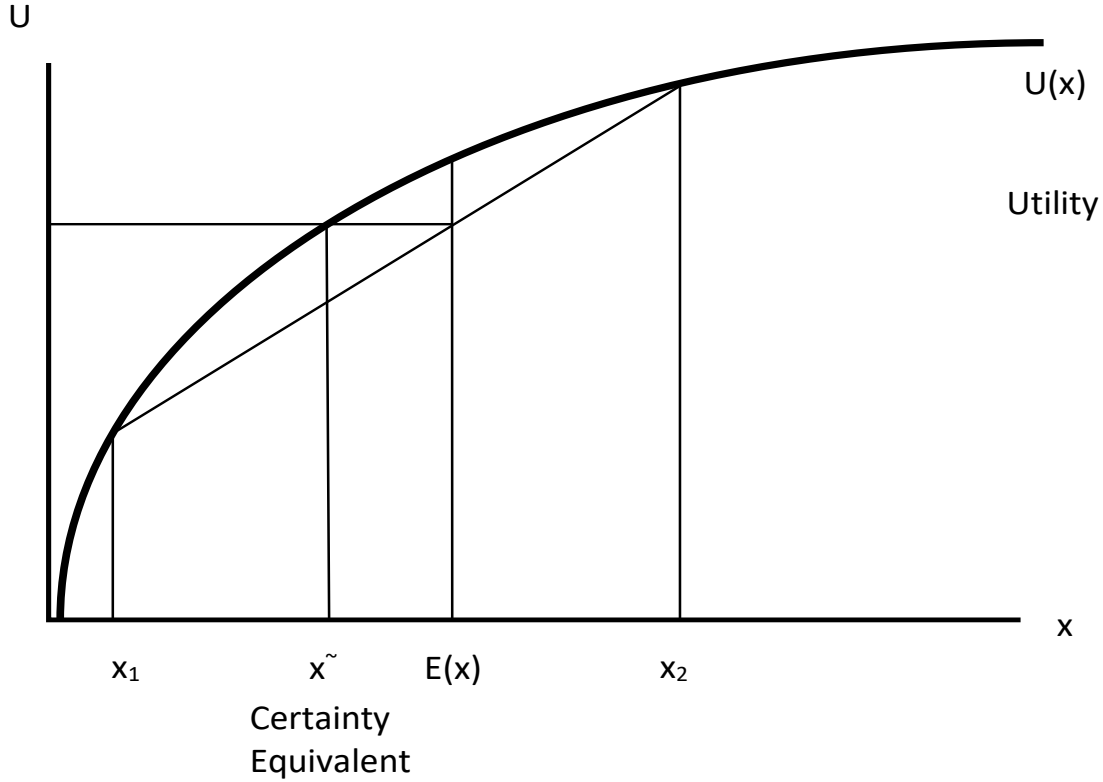
Two possible states of the world x_1, x_2
probabilities p_1, p_2 st $p_1 + p_2 = 1$
E(x)= $x_1 p_1 + x_2 p_2$



Risk aversion: Expected utility is less then utility of expected value.

- **You prefer the sure thing to a fair gamble with the same value**
  - It could be either local (given the lottery)
  - Or global (you always prefer a certain payoff over an uncertain one with the same expected value)
- **There is a price for risk**
  - You have to be paid to bear risk
  - Finance: Return to a security with higher variance is greater than the return to a security with lower variance
- **Risk aversion is generally the rule because people are stuck with what they get.**
  - But suppose you get to experiment: (over jobs, research topics, Vacation locations) should you be risk averse or risk loving?
- **Clean definition of risk-loving: Expected utility is greater than utility of expected value.**
  - Rare for significant investments/positions
  - But people by lottery tickets where the Expected value of a ticket is half what you pay.

What induces you to bear risk? Pricing risk when people are risk averse.



Even if people do not like risk, they will bear it and sometimes they will bear tremendous risk.
They will bear it in particular if they are **paid:** the price of the lottery is less than the certainty equivalent

Certainty Equivalent

- Lottery with expected value is pH +(1-p)L
  - Utility of the expected value U(pH +(1-p)L)
  - Expected Utility is pU(H))+(1-p)U(L).
- What is the certain payoff that gives you the same utility?
  - Find X^C such that $U(X^C)$=pU(H))+(1-p)U(L).
  - If risk averse X^C <pH+(1-p)L

- **Definition**: certainty equivalent is the value of x that given with certainty leaves you indifferent to the gamble between H and L.
  - Certainty equivalent depends on utility (not just return) so it varies across individuals (just like discount rates).
  - Price of the security < certainty equivalent: you buy/hold the security.
  - Price of the security > certainty equivalent: you sell/or do not buy the security.
  - To know how to invest you need to know how patient you are and how risk-averse you are.

From Certainty Equivalent to Insurance

- You own security 1, that pays L if the state is low and H is the state is high. Probability of the state is high is p
- If risk averse $U(pH +(1-p)L) > pU(H))+(1-p)U(L)$.
  - Solution 1 sell the security to someone who is willing to pay at least X^C Someone who is less risk averse
  - Solution 2. buy insurance.
    - Buy a security S_2 that pays y if the state is low and 0 if the state is high.
    - If the seller is risk neutral what is the price of insuring y? $V = y(1-p)$
    - You are fully insured if $H- V =L+y- V$
    - This simplifies to $y=H-L$
    - If you buy this contract you are perfectly insured.

Insurance 2

- We face risk because we ‘own’ the outcome of the security we bought
  - Its payoff is uncertainty
- Insurance is a contingent contract
- We will see other such contracts
  - Futures
  - Options
- Important all these initially designed to distribute risk.

Lessons from today

- Risk aversion is a characteristic of preferences
- You can try to model preferences over individual decisions
  - A trip to Hawaii (weather good or bad)
  - A job (firm does well or not)
  - Lottery ticket
  - Particular asset (a share of So Cal Edison)
- In finance risk aversion is about your total position
  - One trip vs many trips
  - One job (for life) vs moving around
  - One ticket vs many tickets
  - One asset vs a portfolios

Modern finance

- Do not evaluate a security or an action in isolation
- Evaluate its impact on your aggregate wealth position
  - Portfolios matter.
- In fact, you can get insurance for free from a portfolio!